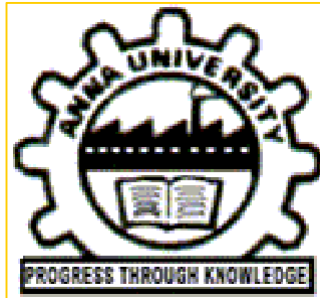


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EC23CO5 ANALOG ELECTRONIC SYSTEM DESIGN

{LAB INTEGRATED THEORY COURSE}

JAN 2025 to MAY 2025

PROJECT REPORT ON

**AUTOMATIC PLANT WATERING SYSTEM
WITH AUDIO ALERT**

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In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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Date:

Signature of the Professor

ABSTRACT

Agriculture, being one of the most essential sectors of the economy, requires efficient and sustainable practices to maximize productivity. One of the major concerns in traditional farming methods is the excessive or insufficient use of water, leading to reduced crop yield and wastage of a precious resource. To address this issue, our project presents a low-cost, automatic irrigation system that monitors soil moisture levels and activates water flow only when necessary, ensuring smart water management.

The system is based on a combination of analog sensing and microcontroller-based control. A soil moisture sensor is used to measure the water content in the soil. This sensor produces an analog voltage corresponding to the moisture level, which is continuously monitored by an Arduino Nano microcontroller. The Arduino is programmed to compare the incoming analog values with a predefined threshold. When the moisture level drops below this set point, the microcontroller sends a digital signal to trigger a relay module. The relay acts as a switch to control a DC water pump or motor, which is turned on to irrigate the soil.

Once the desired moisture level is achieved, the system automatically turns the pump off, preventing over-irrigation. This entire process runs in a closed loop, requiring no manual input, making it ideal for remote or unattended agricultural areas. The use of the Arduino Nano ensures flexibility, compactness, and easy reprogramming if moisture thresholds need to be changed.

This project not only demonstrates the efficient use of analog sensors and digital microcontrollers but also shows the practical application of simple electronic components like relays, transistors, and power supplies in building automation solutions. It can be further enhanced by integrating solar power, wireless monitoring, or expanding the system to support multiple sensor zones.

In conclusion, the automatic irrigation system provides a smart and scalable solution to modern agricultural challenges, helping farmers save time, energy, and water while promoting sustainable farming practices.

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Introduction:

Water is one of the most essential resources in agriculture, and its efficient use is critical for improving crop yield and conserving natural resources. In traditional farming methods, irrigation is usually done manually, which can lead to over-irrigation, under-irrigation, and unnecessary wastage of water. To overcome these challenges, there is a need for a system that can automatically control irrigation based on soil moisture levels.

This project aims to design and implement an Automatic Irrigation System that uses a soil moisture sensor, Arduino Nano, relay module, and DC water pump. The system continuously monitors the soil's moisture level using a sensor. When the moisture drops below a certain threshold, the Arduino Nano processes this data and activates the water pump through a relay. Once the soil reaches the desired moisture level, the pump is automatically turned off.

The use of automation in this project ensures efficient water management, reduces manual labor, and supports better crop health. The Arduino Nano makes the system compact, programmable, and adaptable for future improvements. This project is especially useful for small farms, gardens, and remote agricultural areas where water must be used wisely.

By combining simple electronic components and automation, this project offers a smart and cost-effective solution to modern irrigation needs

Aim:

The aim of this project is to design and implement an Automatic Irrigation System using a soil moisture sensor, Arduino Nano, relay module, and water pump to automate the watering process based on real-time soil moisture levels. The system ensures efficient water usage, reduces manual labor, and supports sustainable agricultural practices by supplying water only when necessary.

Components Required:

S.NO	COMPONENTS	RANGE	QUANTITY
1.	Arduino nano	-	1
2.	Soil moisture sensor	-	1
3.	Relay	5V	1
4.	Buzzer	5V	1
5.	Water pump	9V	1
6.	Bread Board	-	1
7.	Battery	9V	1
8.	Wires	-	As Required

Hardware Description:

1. Arduino Nano

The Arduino Nano is a compact and powerful microcontroller board based on the ATmega328P. It is responsible for processing the input from the soil moisture sensor and controlling the output to the relay module. It has multiple analog and digital pins, making it suitable for sensor interfacing and automation tasks in small-scale projects.

2. Soil Moisture Sensor

This sensor detects the water content present in the soil. It operates by measuring the conductivity between two probes inserted into the soil. Dry soil offers higher resistance, whereas wet soil has lower resistance. The sensor outputs an analog voltage that corresponds to the moisture level, which is read by the Arduino.

3. Relay Module

The relay module acts as a switch to control high-power devices like water pumps using the low-power signal from the Arduino. When the moisture level falls below the threshold, the Arduino sends a HIGH signal to the relay, which closes the circuit and powers the water pump.

4. DC Water Pump

A small DC water pump is used to draw water from a reservoir and irrigate the soil. It operates on 5V–12V DC and is activated through the relay when watering is required.

5. Power Supply

A regulated power supply is used to provide sufficient voltage and current to the Arduino Nano and other components. It may include a battery, adapter, or USB power source depending on the setup.

6. Connecting Wires and Breadboard

These are used to establish connections between all hardware components. A breadboard may be used for prototyping the circuit before final assembly.

Working:

The Automatic Irrigation System works by continuously monitoring the moisture content of the soil using a soil moisture sensor. The sensor detects the level of moisture in the soil by measuring the electrical resistance between two probes. Wet soil has low resistance, resulting in a higher voltage output, while dry soil has higher resistance, which causes a lower output voltage. This analog voltage is sent to the Arduino Nano, which processes the data.

The Arduino Nano compares the sensor's output to a predefined moisture threshold. If the moisture level is above the threshold, the soil is considered sufficiently wet, and no action is taken. However, if the moisture level falls below the threshold (indicating that the soil is too dry), the Arduino sends a signal to the relay module. The relay, acting as a switch, is activated and closes its contacts, allowing current to flow to the DC water pump.

The water pump then draws water from a reservoir and irrigates the soil until the desired moisture level is reached. Once the soil has enough moisture, the sensor detects the change and sends a new signal to the Arduino, which turns off the pump by deactivating the relay. The system continues to monitor the soil's moisture and operates automatically, ensuring that the plants receive water only when necessary. This automated irrigation system optimizes water usage, eliminates the need for manual intervention, and prevents both over-watering and under-watering, making it a cost-effective and sustainable solution for efficient farming and gardening.

CODE:

```
#define DRY_THRESHOLD 400

int soilPin = A0;
int relayPin = 12;    // Motor (Relay)
int buzzerPin = 8;
unsigned long pumpStartTime = 0;
bool motorStarted = false;
bool buzzerChecked = false;
void setup() {
    pinMode(relayPin, OUTPUT);
    pinMode(buzzerPin, OUTPUT);
    Serial.begin(9600);
}
void loop() {
    int moisture = analogRead(soilPin);
    Serial.print("Soil Moisture: ");
    Serial.println(moisture);

    // If soil is dry
    if (moisture > DRY_THRESHOLD) {
        // Start motor if not started
        if (!motorStarted) {
            digitalWrite(relayPin, HIGH);    // Motor ON
            pumpStartTime = millis();        // Start timer
            motorStarted = true;
            buzzerChecked = false;
            Serial.println("Motor ON: Soil is dry");
        }
    }
}
```

```

// After 10 sec, check if still dry
if (!buzzerChecked && (millis() - pumpStartTime >= 10000)) {
    int newMoisture = analogRead(soilPin);
    Serial.print("Recheck Moisture: ");
    Serial.println(newMoisture);

    if (newMoisture > DRY_THRESHOLD) {
        digitalWrite(buzzerPin, HIGH); // Still dry → buzzer ON
        Serial.println("Buzzer ON: No water detected (drought)");
    } else {
        digitalWrite(buzzerPin, LOW); // Moisture improved
        Serial.println("Soil wet after watering: Buzzer OFF");
    }
    buzzerChecked = true;
}
}
// If soil is wet
else {
    digitalWrite(relayPin, LOW); // Motor OFF
    digitalWrite(buzzerPin, LOW); // Buzzer OFF
    motorStarted = false;
    buzzerChecked = false;
    Serial.println("Motor OFF: Soil is wet");
}
delay(500);
}

```

Algorithm:

Step 1: Start the program.

Step 2: Define the threshold and initialize variables.

- Set the dry soil threshold value (e.g., DRY_THRESHOLD = 400).
- Assign the soil moisture sensor to analog pin A0.
- Assign the relay (motor) to digital pin 12.
- Assign the buzzer to digital pin 8.
- Initialize flags like motor Started and buzzer Checked to false.
- Initialize a variable to store the pump start time.

Step 3: Setup the system.

- Set the relay and buzzer pins as output.
- Start serial communication to display real-time data.

Step 4: Continuously read the soil moisture sensor in the loop.

- Read the analog value from the soil moisture sensor.
- Display the moisture value on the Serial Monitor.

Step 5: Check if the soil is dry.

- If the sensor reading is greater than the threshold, the soil is considered dry.

→ If the motor is not already running:

- Turn ON the motor (activate the relay).
- Record the current time using millis().
- Mark the motor as started.
- Reset the buzzer Checked flag to false.
- Print message: "Motor ON: Soil is dry"

→ If 10 seconds have passed since motor started and buzzer is not yet checked:

- Read the moisture sensor again.
- If the soil is still dry:
 - Turn ON the buzzer to indicate a problem (e.g., water not reaching roots).

- Print message: "Buzzer ON: No water detected (drought)"
- Else:
 - Turn OFF the buzzer.
 - Print message: "Soil wet after watering: Buzzer OFF"
- Set buzzer Checked to true.

Step 6: If the soil is wet ($\text{moisture} \leq \text{threshold}$):

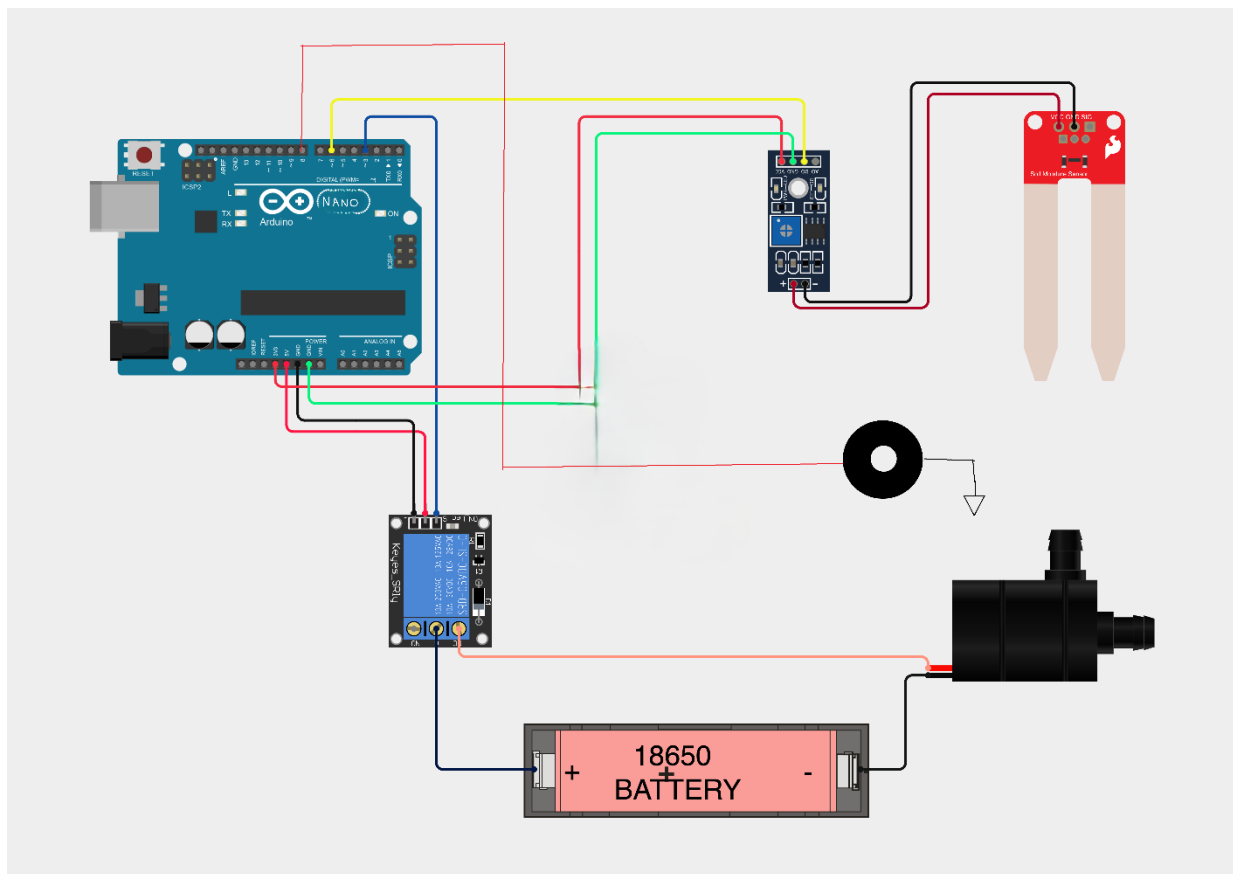
- Turn OFF the motor.
- Turn OFF the buzzer.
- Reset the flags motor Started and buzzer Checked to false.
- Print message: "Motor OFF: Soil is wet"

Step 7: Wait for a short delay (e.g., 500 milliseconds).

Step 8: Repeat the loop from Step 4.

Circuit Diagram :

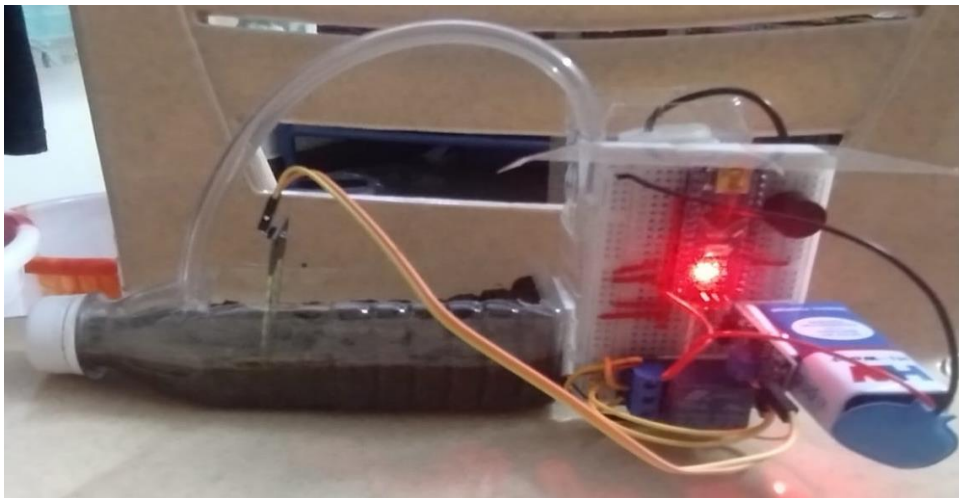
Circuit diagram:



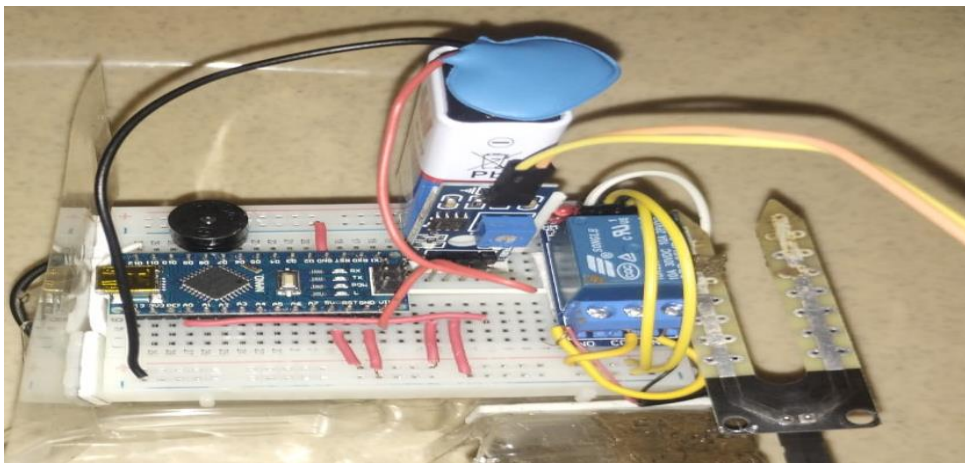
Experimental setup:



Activation of circuit:



Circuit connection:



Applications:

- The system automates irrigation for small-scale farms, improving water efficiency and crop yield.
- It reduces manual effort in home gardens by ensuring plants are watered only when needed, preventing over or under-watering.
- It maintains consistent soil moisture in greenhouses, supporting plant growth while minimizing labour costs.
- The system automates irrigation for lawns and public parks, conserving water and reducing maintenance efforts
- It provides an efficient irrigation solution for remote or unmanned areas, ensuring crops or plants are watered consistently without human supervision.

Advantages:

- **Water Conservation:**
Efficiently uses water by irrigating only when needed, reducing waste.
- **Automation:**
Operates autonomously, saving time and effort.
- **Cost-Effective:**
Lowers water bills and is affordable to implement.
- **Improved Plant Growth:**
Prevents over or under-watering, promoting healthier plants.
- **Energy Efficient:**
Minimizes energy consumption by activating the pump only when required.
- **Remote Monitoring:**
Ideal for remote areas, requiring no constant supervision.

Conclusion:

The Automatic Irrigation System provides an efficient and sustainable solution to the challenges of traditional manual irrigation. By utilizing a soil moisture sensor, Arduino Nano, relay module, and a water pump, the system is able to monitor soil moisture levels and automatically control irrigation, ensuring that plants receive water only when necessary. This not only conserves water but also ensures optimal plant growth by preventing the issues of over-watering and under-watering.

The use of an Arduino Nano allows for simple programming and customization, making the system scalable and adaptable for various applications, from small home gardens to large-scale farms. The integration of a soil moisture sensor ensures real-time data collection, enabling precise irrigation, while the relay module effectively controls the water pump based on the sensor's input.

In addition to its practical applications in agriculture, home gardening, and greenhouses, this system offers numerous advantages, including water and energy conservation, automation, and cost savings. Furthermore, the system is ideal for remote areas, reducing the need for manual intervention and enabling effective irrigation even in the absence of regular supervision.

The project also opens up possibilities for future enhancements, such as wireless control, scheduling features, or integration with weather data for more intelligent irrigation management. Overall, this Automatic Irrigation System offers a smart, sustainable, and cost-effective solution for modern agriculture and gardening needs, addressing both environmental concerns and the growing demand for efficient resource management.

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